

$$v = 4.5 - 1.5 = 3.0 \text{ m/s}$$

7	D	Since cube is floating, there is vertical equilibrium. $U_1 + U_2 = W_{\text{cube}}$ $V_1 \rho_1 g + V_2 \rho_2 g = (V_1 + V_2) \rho_c g$ $V_1 \rho_1 + V_2 (3\rho_1) = (V_1 + V_2) (2\rho_1)$ $3V_2 - 2V_2 = 2V_1 - V_1$ $V_2 = V_1$ $\frac{V_1}{V_2} = 1$
8	A	Work done by the force to extend the spring is given by the area under force-extension